

01 Problem Statement

Who is affected, and why it matters

East Africa's mobile money networks — MTN MoMo, M-Pesa, Airtel Money, Tigo — already move value across borders. But every rail in use today was built for **personal remittances**: a person sending money home to family. None were built for **business trade**. That single design gap is what a Kigali trader paying a Nairobi supplier runs into, every time:

- **Trade-sized limits are missing** — caps near USD 1,700 sit far below a real bulk trade payment.
- **No invoice link** — a payment is just "money moved"; neither side has a record of what it paid for, so disputes default to one word against another.
- **No business/personal separation** — one wallet does everything, so bookkeeping, tax prep, and loan applications become guesswork.
- **No visible trust history** — a new supplier cannot verify a buyer, so traders are forced into 50–60% upfront deposits before goods ship.
- **Blind USSD interface** — no live confirmation, no visible exchange rate, no reviewable history.

This falls on the East African SME trader — importers, exporters, and their suppliers — who are running a real business on infrastructure that was never designed to carry one. The cost is measured in cash tied up in deposits, disputes with no record to settle them, and a credit history that never gets built.

02 Innovation & Differentiation

What's genuinely new, and why usage grows

VUKA is not a new payment rail competing for share against MTN, M-Pesa, or continental infrastructure. It is the **trader-facing layer that has never been built on top of them** — the missing front door, not a new road.

Existing player	What it offers	What it does not offer
MTN MoMo / M-Pesa direct corridor	Basic wallet-to-wallet transfer	No invoicing, no trust history, no bookkeeping separation, no interface
Onafriq (formerly MFS Africa)	Backend interoperability across markets	No trader-facing application — partners build their own front end on top
PAPSS	Continental bank settlement infrastructure	Not built for informal SME traders on mobile money

VUKA adds the four things none of the above provide — invoice-linked payments, engine-enforced business/personal separation, visible payment-history trust, and a real confirmable interface. This directly removes the two frictions traders report most: paying a large deposit because a new supplier can't verify them, and being unable to reconcile what a payment was even for. Usage compounds by design — the longer a trader uses VUKA, the more trust history they carry, and the smaller the deposit their next supplier will ask for.

03 Readiness for Testing & Market Deployment

What is built, what is needed

BUILT & VERIFIED TODAY

- ✓ PostgreSQL double-entry ledger — balances always derived, never stored
- ✓ Go transfer engine, full unit test coverage, safe concurrency
- ✓ Working telecom adapter seam (MTN/M-Pesa), simulated end-to-end
- ✓ Business/personal separation enforced at the engine level
- ✓ React dashboard — live confirmation, history, status tracking

WHAT WE'RE ASKING BNR FOR

- Guidance through the BNR fintech sandbox application
- Introductions to a telecom or aggregator partner (e.g. Onafriq) for real rail access
- Regulatory feedback on an invoice-linked, business-focused product on mobile money rails
- Rwanda's active passporting agreements (Ghana, Feb 2025 · Kenya, Mar 2026) already open the legal pathway

Verification record (final build): go vet ✓ · engine test suite passing (ledger, api) · tsc 0 errors · 37/37 frontend unit tests · production build passing · ledger integrity confirmed — every double-entry transfer nets to zero across a full live demo dataset.

04 Expected Impact

What changes for the trader, and how it scales

Disputes Every payment is tied to an invoice number both sides hold — no more "I paid" versus "you never did."	Bookkeeping Business income and expense are separated from personal spend from day one — audit and tax-ready.	Access to capital A visible, verifiable on-time payment history becomes the foundation for future lender and trade-finance products.
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Every corridor added after the pilot reuses the same engine against a new telecom pair — the cost of expansion is a new adapter, not a new product. Longer term, the accumulated ledger of on-time payments becomes a credit signal VUKA can expose to lenders directly: a trade-finance on-ramp for SMEs who today have no credit history at all.

05 User Journey

Amina, a Kigali coffee importer, paying Kethan, a Nairobi supplier

- 1 Onboard** — Amina registers on VUKA and automatically receives two wallets, PERSONAL and BUSINESS, structurally separate from the start.
- 2 Receive an invoice** — Kethan issues an invoice tied to a specific order: line items, VAT, due date. Amina sees it inside VUKA, not a text message she has to take on trust.
- 3 Review before paying** — Amina sees the live exchange rate and full confirmation details before committing — no blind USSD menu.
- 4 Pay** — Amina pays from her BUSINESS wallet, against the invoice number. The system will not allow payment from her personal wallet or against an undocumented invoice.
- 5 Both sides see the same fact** — the invoice flips to PAID for Amina and Kethan simultaneously, derived from one shared ledger — not reported separately by either side.
- 6 Trust compounds** — that completed, on-time payment joins Amina's visible track record, letting her negotiate a smaller deposit the next time she orders from someone new.

06 Payment Process Flow

Initiation to settlement, and every party involved

1. Initiate Payer selects an invoice or transfer and confirms amount and live FX rate in the VUKA dashboard.	2. Process Engine writes one idempotent transaction. A cross-border payment writes two currency legs — four linked ledger entries — inside a single commit.	3. Confirm If the telecom rail succeeds, entries commit and invoice status is derived as PAID; any failure rolls the whole transfer back — no partial state.	4. Complete Payer and payee both see the same settled record from their own side of the ledger — one row, two seats, fully traceable.
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Example — RWF → KES cross-border transfer (excerpt, must net to zero)

debit	Amina BUSINESS (RWF)	-190,000 RWF
credit	VUKA Settlement (RWF)	+190,000 RWF
debit	VUKA Settlement (KES)	-20,000 KES
credit	Kethan BUSINESS (KES)	+20,000 KES

Parties involved: the payer (trader), the payee (supplier), the telecom rail — MTN MoMo or M-Pesa, which mediates the actual movement of funds so VUKA never holds a MoMo PIN — and the VUKA engine/ledger, the single source of truth both sides read from.